THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — SCIENCE · HARD LEVEL CHAPTER TEST SET 3

Chapter 12: **How Nature Works in Harmony** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Which of these may lead to the extinction of species? 1

(a) habitat destruction **(b)** conservation laws
(c) afforestation **(d)** awareness campaigns

2. Assertion (A): Legume plants help in making the soil fertile. Reason (R): Rhizobium in their root nodules fixes atmospheric nitrogen. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

3. A dead log in a forest is full of fungi and insects. This shows: 1

(a) decomposition and recycling of nutrients **(b)** photosynthesis
(c) pollination **(d)** predation only

4. Rainwater harvesting increases: 1

(a) groundwater level **(b)** soil erosion
(c) global warming **(d)** air pollution

5. Assertion (A): Deforestation increases the CO2 concentration in the atmosphere. Reason (R): Trees take in CO2 during photosynthesis. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

6. If all bees disappeared, which process would suffer most? 1

(a) respiration **(b)** pollination of flowering plants
(c) decomposition **(d)** evaporation

7. Zero-waste practices mean: 1

(a) burning waste
(b) burying waste
(c) reducing waste by reuse, recycling and composting
(d) dumping waste

8. Assertion (A): Wolves were reintroduced into a national park to restore balance. Reason (R): Wolves control the numbers of herbivores like elk. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

9. If all producers die in an ecosystem, then: 1

(a) decomposers take over
(b) consumers become producers
(c) all consumers eventually die
(d) nothing changes

10. A sustainable development goal is to: 1

(a) use all resources now
(b) stop all development
(c) build on forests
(d) meet present needs without harming the future

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Why does a decrease in dissolved oxygen of water cause fish to die? 2

12. State why green plants in a pond are more in number than fish. 2

13. Why is the use of biodegradable bags preferred? 2

14. What is meant by the term 'sustainable use of resources'? Give one example. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Why are wetlands, mangroves and coral reefs called hotspots of biodiversity? 3

16. How do the Bishnoi people of Rajasthan protect the environment? Mention two practices. 3

17. Why are national parks and wildlife corridors important for large animals? 3

OR

Explain why a balanced aquarium can survive for a long time with plants, fish and snails. 3

18. Explain how a food web helps to conserve species in a protected area. 3

OR

Explain how crop residues can be used sustainably instead of burning. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Explain how the tiger reserve project and the Chipko movement represent institutional and community action respectively. 5

OR

A town has no trees, no ponds and many concrete roads. Describe the effects on temperature, air quality, water table and biodiversity and suggest five changes. 5

20. Plan an experiment to demonstrate that photosynthesis produces oxygen in a water plant and that decomposers need oxygen in a compost heap, including what you will observe. 5

OR

Explain sustainable development using an example of a village that manages its forest, water and waste well. 5

END OF QUESTION PAPER — Total: 40 Marks